Lab session 5

Target of the session : Canny edge detector

Algorithm steps:

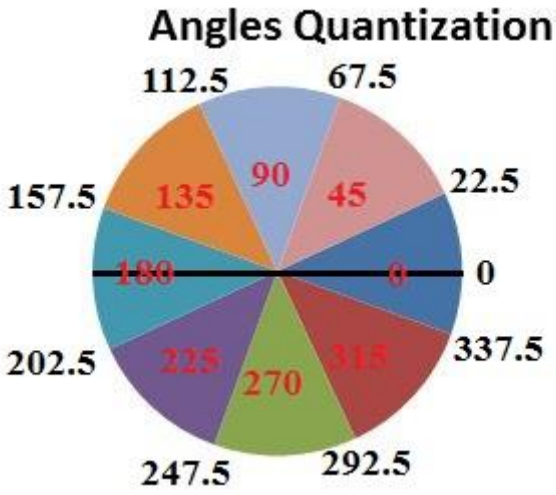
1. Resize the input gray image to become of the size [200, 200]: just for speeding up the execution time of the algorithm. (optional)
2. Remove the noise from the input gray image using [5.5] Gaussian kernel and sigma = 1.
3. Apply Sobel edges detector using two masks and convolution operation to calculate the first order derivatives (Ix, Iy), and calculate the Magnitude (E) and angles of the detected edges.
   (Make sure that E in the range [0, 1] and the angles are positive and in degrees not in radian.)
4. Quantize the angles to belong to the set {0, 45, 90, 135, 180, 225, 270, 315}
5. Eliminate the repeated edges from (E) using non-maximum suppression process in the edge direction.
6. Classify the edges in (E) into {Strong, Candidate, weak} by choosing two threshold. keep the strong edges. Eliminate weak edges.
7. Eliminate Candidate edges that aren't connected to any strong edges in any direction.

Some explanation:

Sobel Kernels:

$$Hx = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, Hy = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

$$E = \sqrt{Ix^2 + Iy^2}\,, angles\ (in\ radian) = arctan2(Iy, Ix)$$



Non-Maximum suppression

Choose two neighbors of the current pixel (i, j) from (E) according to its angle:
If the value of E(i, j) is greater than its neighbors, then keep it, else eliminate it.

Final Results of the session:

Image	Max threshold	Min threshold
City	0.3	0.03
I2	0.1	0.02
I3	0.23	0.03

